

Asking “Which Is Worse?” Barely Changes Which Response Four Sub-2B Qwen Judges Prefer: A Letter-Calibrated Negation Test on a Code-Heavy RewardBench Subsample

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Abstract—Small open language models are attractive as cheap pairwise judges, but does their verdict depend on the criterion asked? We test this by negation: each pair is judged under “Which response is better?” and “Which response is worse?”, in both presentation orders. We evaluate four Qwen judges (Qwen2.5-0.5B/1.5B-Instruct, Qwen3-0.6B/1.7B; thinking off; next-token A-vs-B logit readout; CPU) on 300 RewardBench pairs, 217 of which are HumanEvalPack code pairs. Under the argmax readout, a correct reversal occurs in 0, 0, 0 and 1 of 300 pairs. Our pre-registered “position-locking” hypothesis (same letter in all four judgments) is supported for the two smallest judges (1.000 and 0.890), but a letter-biased null shows this is largely implied by their letter rates. A post-hoc letter-calibrated analysis that averages the A-minus-B logit over both orders shows the actual failure: the pair-dependent logit shift under “worse” is strongly positively correlated with the shift under “better” (Pearson $r = 0.95, 0.63, 0.90, 0.63$; same sign in $0.71\text{--}0.92$ of pairs), so the question barely changes which response is favored. The 0.95 comes from Qwen2.5-0.5B, which always answers “A” and whose shift is tiny relative to its letter offset; the three other judges give $0.63\text{--}0.90$. Only Qwen3-1.7B shows clear content signal (calibrated “better” accuracy 0.81 vs. $0.50\text{--}0.61$), yet it does not reverse. The pattern holds on the 83 non-code pairs ($r\ 0.57\text{--}0.95$) and on high-format-compliance pairs. Generation-based readouts, thinking, and judges above 2B were not tested.

Index Terms—LLM-as-a-judge, pairwise evaluation, criterion negation, position bias, calibration, small language models, negative result

I. INTRODUCTION

Pairwise LLM judges are widely used to compare model outputs [1], and their position bias is well documented [2], [3], [4]. Small open models (below 2B parameters) are attractive judges because they run on commodity CPUs, and recent work evaluates Qwen3 models as small as 0.6B as judges on RewardBench [5]. A basic precondition for any judge is that its verdict depends on the criterion it is asked about. Accuracy on a “which is better?” prompt cannot establish this, and an order-swap test detects position bias but not whether the judge reads the question.

We probe criterion conditioning by negation, a property known as negation invariance in the logical-consistency literature [6]: a judge that prefers the chosen response under “better” should prefer the rejected one under “worse”. Asking both

questions in both orders gives four judgments per pair. Our first analysis, which was pre-registered, classified each pair by the four argmax verdicts and concluded that the smallest judges are “position-locked”. Referee feedback showed that this conclusion was driven by the readout: strongly letter-biased judges produce the same letter four times regardless of what they prefer. This revision therefore separates the letter prior from content preference by averaging over presentation orders, and finds a pair-dependent shift that is mostly the same under both questions.

Contributions.

- 1) A four-judgment negation design with two readouts: a pair-level argmax taxonomy and a letter-calibrated order-averaged logit difference that removes a constant letter offset (Section III).
- 2) An evaluation of four sub-2B Qwen judges (thinking off) on 300 RewardBench pairs with pinned model and data revisions (Section IV).
- 3) The finding, for these judges and this sample, that the order-averaged logit shift under “worse” is strongly positively correlated with that under “better” (criterion-blindness), robust to excluding code pairs and low-mass passes (Section V).
- 4) A correction of our own pre-registered framing: under a letter-biased null the position-locking hypothesis is nearly implied by letter marginals, so argmax taxonomies can misattribute criterion-blindness to position bias (Sections V and VI).

II. RELATED WORK

Negation and logical consistency of judges. Liu et al. [6] propose measuring logical preference consistency of LLM judgments by transitivity, commutativity and negation invariance (whether a model inverts its judgment when the relation is negated), evaluate open models from Phi-3-mini to 13B and two API models, and introduce the REPAIR data augmentation. Our “better” vs. “worse” test is an instance of negation invariance; we add order swaps, sub-2B judges and a letter-calibrated readout. GRP [7] asks strong closed judges which answer is *worse* as a prompting method and reports improved consistency on JudgeBench. Bagaria et al. [8] reverse the

rubric criterion in pointwise judging with Qwen2.5-7B-Instruct and report desired verdict flips of 16.8% and 32.2% (two candidate models) under criterion reversal. PRIN [9] contrasts “which are correct?” with “which are incorrect?” in multiple-choice settings with models of 8B parameters and larger. Elkins and Chun [10] audit negated vs. affirmative framings of moral dilemmas in 16 models and report swings of up to 76 percentage points for small (1–4B) models; their task is stance-taking, not response comparison. JudgeSense [11] measures judge sensitivity to prompt rewording more broadly.

Position bias and hidden signal. Zheng et al. [1] report position and verbosity bias; Wang et al. [2] propose balanced-position calibration, which our order-averaged readout resembles; Shi et al. [3] study position bias across 15 judges; Norman et al. [4] evaluate 21 judges from nine providers and report high test-retest reliability coexisting with severe position bias in two production-deployed judges. Mukherjee and Sitaram [12] study 64 open-weight judges and report that on LLMBar [13] the verdicts of 50 judges agree with human labels only 0.456 of the time even after averaging both orders, while probes on activations recover much more signal. Our calibrated analysis makes a related, simpler observation at the output layer: the argmax verdict of a letter-biased judge hides a content preference that the logit difference still carries.

Negation in language models. Pretrained LMs often fail to distinguish negated from affirmative probes [14], and larger LMs can perform worse on negated prompts [15].

Difference from prior work. Negation invariance of judges [6] and reversed-goal prompting [7] are established ideas, so our contribution is narrow: applying the negation test in both orders to four judges below 2B parameters, and showing that a letter-calibrated readout changes the diagnosis from position-locking to criterion-blindness. Our literature scan used arXiv and web search only; we do not claim to be first.

III. METHOD

Prompt. Each judgment uses the model’s chat template with a single user turn containing the user request, “[Response A]” and “[Response B]”, a question, and “Answer with a single letter: A or B.” The questions are *better* (“Which response is better?”), *W1* (“Which response is worse?”, primary negation), *W2* (“Which response should be rejected?”) and *length* (“Which response is longer?”, control). For Qwen3 we pass `enable_thinking=False`.

Readout. One forward pass; the score s is the difference of next-token logits of the single tokens “A” and “B”; the argmax verdict is the letter with the higher logit. The probability mass on {“A”, “A”, “B”, “B”} is recorded as a format-compliance measure. Inference is deterministic.

Four judgments per pair. For each pair (chosen C , rejected R) we ask *better* and *W1*, each with C shown as A (order *cf*) and with R shown as A (order *rf*).

Argmax taxonomy (pre-registered). Each pair receives the first matching category: (1) *correct reversal*: C in both orders

TABLE I
JUDGES AND DATA WITH PINNED REVISIONS (FIRST 12 HEX DIGITS; FULL SHAS IN RESULTS/MODELS.MD).

Repository	Revision	Params	License
Qwen/Qwen2.5-0.5B-Instruct [16]	7ae557604adf	494,032,768	apache-2.0
Qwen/Qwen2.5-1.5B-Instruct [16]	989aa7980e4c	1,543,714,304	apache-2.0
Qwen/Qwen3-0.6B [17]	c1899de289a0	751,632,384	apache-2.0
Qwen/Qwen3-1.7B [17]	70d244cc86cc	2,031,739,904	apache-2.0
allenai/reward-bench (data) [18]	168d848cddbbe	–	odc-by

under *better* and R in both orders under $W1$; (2) *position-locked*: the same letter in all four judgments; (3) *criterion-blind*: the same response in all four; (4) *reversed-consistent*; (5) *other*. Under a uniformly random judge the shares are 1/16, 2/16, 2/16, 1/16 and 10/16. *Position-consistent accuracy* is the share of pairs with C chosen in both orders under *better* (the conditioning subset, size n_{cond}); the *conditional flip rate* is the share of that subset for which $W1$ selects R in both orders.

Letter-calibrated readout (post hoc). For each pair and question q , $d_q = (s_q^{cf} - s_q^{rf})/2$. If the judge adds a constant offset to the “A” logit, it cancels; $d_q > 0$ means the judge favors C as the answer to q . A criterion-following judge has $d_{\text{better}} > 0$ and $d_{W1} < 0$ on most pairs, i.e. a negative correlation; a criterion-blind judge, which favors the same response whatever the question, has a positive correlation. We report Pearson $r(d_{\text{better}}, d_{W1})$, the same-sign share, calibrated accuracy (share with $d_{\text{better}} > 0$) and calibrated reversal (share with $d_{\text{better}} > 0$ and $d_{W1} < 0$), with 95% bootstrap CIs over pairs (2000 resamples, seed 0).

Letter-biased null. For each judge, an independent-letter judge with the observed per-question “A” rates p_b (better) and p_w ($W1$) gives all four judgments the same letter with probability $p_b^2 p_w^2 + (1 - p_b)^2 (1 - p_w)^2$ and has position-consistent accuracy $p_b(1 - p_b)$.

Hypotheses (pre-registered). H1: the conditional flip rate rises with size within each family (paired bootstrap, Bonferroni-adjusted 97.5% CIs); declared inconclusive if any $n_{\text{cond}} < 30$. H2: among non-correct-reversal pairs, the position-locked share exceeds 0.5 (95% CI lower bound > 0.5), for Qwen2.5-0.5B and Qwen3-0.6B. The calibrated readout and letter-biased null were added after review and are not pre-registered. A development-set A/B-mass exclusion check (< 0.5) was run for Qwen2.5-0.5B only (20 pairs, mean mass 0.9998), a protocol deviation; on the evaluation set all judges have mean mass ≥ 0.745 .

IV. EXPERIMENTAL SETUP

Models. Four judges loaded from safetensors with `trust_remote_code=False` in float32 (Table I); all Apache-2.0 per their model cards.

Data. RewardBench [18] (ODC-By) at revision 168d848cddbba9764fae4a544dc9ca1e6cca4931, sections chat, chat-hard and reasoning. Of 2985 items we excluded 740 in safety subsets and 1 matching a safety keyword list, 1021 with a response longer than 150 tokens, and 213 whose full prompt exceeded 400 tokens, leaving

1010 (chat 46, chat-hard 275, reasoning 689). We drew 30 development pairs (format checks only) and three disjoint resamples of 100 evaluation pairs (sampling seeds 0, 1, 2), pooled to 300: chat 12, chat-hard 65, reasoning 223. **Composition caveat:** the length filter makes the sample code-heavy: 217 of 300 pairs are HumanEvalPack pairs (correct vs. buggy code for the same task, often near-identical text; we did not quantify textual overlap), and 83 are other pairs. We report results on both subsets.

Passes. Per model, 300 pairs $\times$ 4 judgments plus, on resample 0 only, 100 pairs $\times$ 2 orders for W2 and for the length control: 1600 passes per model. Inference is deterministic, so the resamples are data splits, not seeds; there is one pass per prompt and no seed variance. Uncertainty is quantified by bootstrap over pairs. The calibrated analysis reuses the same raw outputs; no new model runs.

Hardware and budget. CPU only: one shared aarch64 machine, 4 threads, float32; Python 3.14.4, torch 2.14.1, transformers 4.57.6 (results/pip_freeze.txt). Summed per-pass forward time is 2.37 h (879 s, 3281 s, 1153 s, 3222 s). Extension judges (Qwen2.5-3B, Llama-3.2-1B) were allowed only if the core sweep was projected at ≤ 1.5 h; the projection was 2.32 h, so none was run.

V. RESULTS

Argmax readout (pre-registered). Table II gives the pooled taxonomy. Correct reversal is 0.000, 0.000, 0.000 and 0.003 (1 of 300 pairs, Qwen3-1.7B), below the random-judge 0.062. Position-consistent accuracy under *better* is 0.000 to 0.430. H2 is supported for both pre-specified judges: the position-locked share of non-correct-reversal pairs is 1.000 [1.000, 1.000] for Qwen2.5-0.5B, which answers “A” on all 1600 passes (its row equals the always-A reference), and 0.890 [0.857, 0.927] for Qwen3-0.6B (exploratory: Qwen2.5-1.5B 0.710 [0.657, 0.763], Qwen3-1.7B 0.431 [0.378, 0.487]). H1 is inconclusive by the pre-registered rule ($n_{\text{cond}} = 0$ and 15 for the small judges); the Qwen3 conditional flip rate moves from 0.000 to 0.008 (97.5% CI of the difference [0.000, 0.030]), and the chance-normalized unconditional flip differences have 97.5% CIs [0.000, 0.040] (Qwen2.5) and [−0.018, 0.053] (Qwen3), so there is no evidence of a size trend in this range.

H2 is close to tautological. The letter rates explain most of the argmax picture. Per-question “A” rates (better/W1) are 1.00/1.00, 0.20/0.01, 0.94/0.94 and 0.34/0.12 (Fig. 1). An independent-letter judge with these rates gives all four judgments the same letter in 1.00, 0.63, 0.79 and 0.34 of pairs; observed: 1.00, 0.71, 0.89 and 0.43 (Table III). Against the uniform random judge (0.125) H2 is therefore nearly guaranteed for any strongly letter-biased judge, and its support tells us about letter bias, not about the criterion. Likewise, position-consistent accuracy under the letter-biased null is 0.000, 0.159, 0.053 and 0.224, close to the observed 0.000, 0.103 and 0.050 for the three smaller judges; only Qwen3-1.7B (0.430) exceeds what its letter rates predict.

Calibrated readout: criterion-blindness. Table III and Fig. 2 show the letter-calibrated analysis. For all four judges,

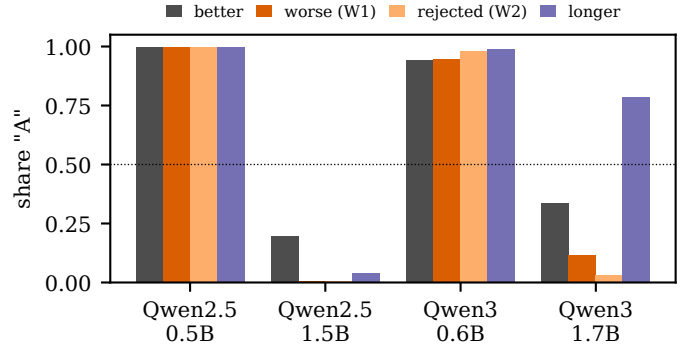


Fig. 1. Share of passes answered “A” per question (better and W1: 600 passes per judge; W2 and length: 200 passes, resample 0 only). Dotted line: 0.5.

d_{W1} is strongly *positively* correlated with d_{better} : $r = 0.95$ [0.92, 0.97], 0.63 [0.51, 0.73], 0.90 [0.87, 0.93] and 0.63 [0.55, 0.71]. The same sign occurs in 0.92, 0.71, 0.87 and 0.76 of pairs, and every CI lower bound is above 0.5 (0.88, 0.66, 0.82, 0.71). A criterion-following judge would show a negative correlation. Thus, once the letter offset is removed, each pair still produces a shift d whose sign is mostly unchanged by the question; the question mainly moves the letter prior. This shift need not be a quality judgment: calibrated “better” accuracy is 0.61, 0.50, 0.57 and 0.81, i.e. near chance for the three smaller judges, and only Qwen3-1.7B shows clear content signal under *better*, which it does not invert under W1. For Qwen2.5-0.5B, d is also tiny compared with the letter offset: the median $|d_{\text{better}}|$ is 0.047 logits against a mean order-averaged A-minus-B score of 2.80 (Qwen3-1.7B: median $|d_{\text{better}}|$ 5.4). Its $r = 0.95$ therefore shows that small residual shifts are reproduced across questions, not that this judge has a coherent preference between the responses. The calibrated-reversal shares (0.06 to 0.22) are the tail where d changes sign; we defined no null for them and do not interpret them as evidence of reversal. Under the argmax taxonomy the three smaller judges look position-locked rather than criterion-blind (criterion-blind shares 0.000 to 0.113 for all four judges in Table II) because a large letter offset dominates the argmax verdict.

Robustness. (i) *Sample composition:* the correlation is 0.57–0.95 on the 83 non-HEP pairs and 0.58–0.96 on the 217 HEP pairs (Table III). Calibrated “better” accuracy is lower on non-HEP pairs (0.43, 0.53, 0.48, 0.61) than on HEP pairs (0.68, 0.48, 0.61, 0.88), so the content signal of Qwen3-1.7B is stronger on code pairs, but criterion-blindness is present in both subsets. (ii) *Format compliance:* restricting to pairs with A/B mass > 0.9 on all four passes ($n = 300, 288, 286, 140$) gives $r = 0.95, 0.60, 0.91, 0.63$. For Qwen3-1.7B, whose mean mass is 0.745 and 25.2% of whose better/W1 passes have mass below 0.5, this filter keeps less than half of the pairs. (iii) *Not done:* we did not check the logit readout against generated verdicts.

Controls (resample 0, 100 pairs, exploratory, argmax). With W2 (“Which response should be rejected?”) the correct-

TABLE II

ARGMAX READOUT, POOLED OVER 300 REWARDBENCH PAIRS PER JUDGE (W1 = “WHICH RESPONSE IS WORSE?”). SHARES OVER ALL 300 PAIRS. n_{cond} : PAIRS CORRECT UNDER “BETTER” IN BOTH ORDERS. REFERENCE ROWS ARE ANALYTIC. “UNDEF.”: $n_{\text{cond}} = 0$. POSITION-LOCKED SHARES ARE LARGELY PREDICTED BY LETTER RATES (TABLE III); SEE TEXT.

Judge	n_{cond}	Acc. (better)	Cond. flip	Uncond. flip	Correct rev.	Pos.-locked	Crit.-blind	Rev.-consist.	Other	A/B mass
Qwen2.5-0.5B	0	0.000	undef.	0.000	0.000	1.000	0.000	0.000	0.000	1.000
Qwen2.5-1.5B	31	0.103	0.000	0.013	0.000	0.710	0.003	0.000	0.287	0.996
Qwen3-0.6B	15	0.050	0.000	0.027	0.000	0.890	0.023	0.000	0.087	0.975
Qwen3-1.7B	129	0.430	0.008	0.040	0.003	0.430	0.113	0.000	0.453	0.745
Random judge	–	0.250	0.250	0.250	0.062	0.125	0.125	0.062	0.625	–
Always-A judge	0	0.000	undef.	0.000	0.000	1.000	0.000	0.000	0.000	–

TABLE III

LETTER-CALIBRATED ANALYSIS (POST HOC), $d_q = (s_q^{cf} - s_q^{rf})/2$. r : PEARSON CORRELATION OF d_{better} AND d_{W1} (CRITERION-FOLLOWING $\Rightarrow r < 0$; CRITERION-BLIND $\Rightarrow r > 0$). SAME SIGN: SHARE WITH $\text{sign}(d_{\text{better}}) = \text{sign}(d_{W1})$. CAL. ACC.: SHARE WITH $d_{\text{better}} > 0$. CAL. REV.: SHARE WITH $d_{\text{better}} > 0$ AND $d_{W1} < 0$ (NO NULL DEFINED). HEP: HUMANEVALPACK. MASS>0.9: PAIRS WITH A/B MASS > 0.9 ON ALL FOUR PASSES. LETTER NULL: SHARE OF PAIRS WITH THE SAME LETTER IN ALL FOUR JUDGMENTS, EXPECTED UNDER INDEPENDENT LETTERS AT OBSERVED RATES / OBSERVED. 95% BOOTSTRAP CIs IN BRACKETS.

Judge	All pairs ($n = 300$)				Non-HEP ($n = 83$)	HEP ($n = 217$)	Mass>0.9	Letter null
	r	Same sign	Cal. acc.	Cal. rev.				
Qwen2.5-0.5B	0.95 [0.92, 0.97]	0.92 [0.88, 0.95]	0.61	0.06 [0.03, 0.08]	0.95 [0.92, 0.97]	0.96 [0.94, 0.97]	300; 0.95 [0.92, 0.97]	1.00 / 1.00
Qwen2.5-1.5B	0.63 [0.51, 0.73]	0.71 [0.66, 0.76]	0.50	0.22 [0.18, 0.27]	0.72 [0.61, 0.82]	0.58 [0.38, 0.72]	288; 0.60 [0.47, 0.71]	0.63 / 0.71
Qwen3-0.6B	0.90 [0.87, 0.93]	0.87 [0.82, 0.90]	0.57	0.08 [0.05, 0.11]	0.91 [0.88, 0.94]	0.91 [0.84, 0.95]	286; 0.91 [0.88, 0.94]	0.79 / 0.89
Qwen3-1.7B	0.63 [0.55, 0.71]	0.76 [0.71, 0.81]	0.81	0.18 [0.14, 0.23]	0.57 [0.38, 0.72]	0.69 [0.59, 0.77]	140; 0.63 [0.50, 0.73]	0.34 / 0.43

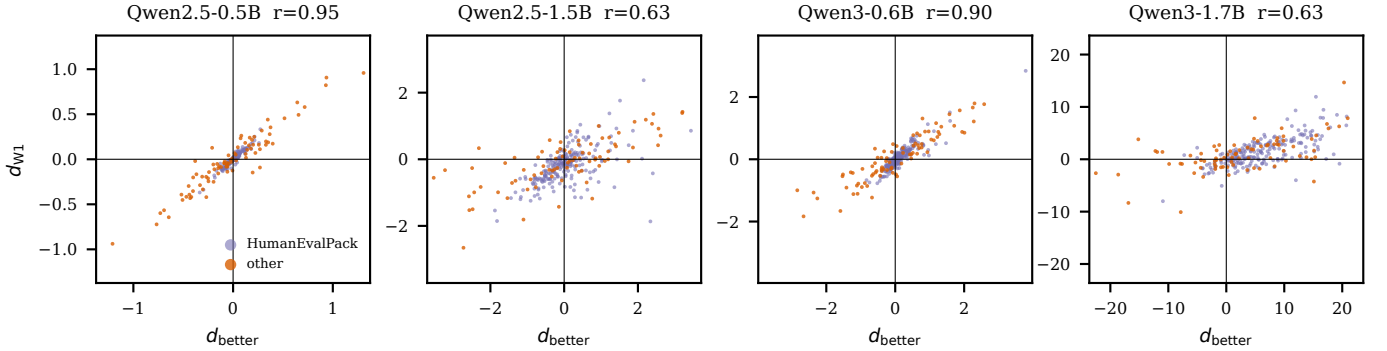


Fig. 2. Letter-calibrated preference for the chosen response under “better” (x) vs. “worse” (y), one point per pair ($n = 300$ per judge; purple: HumanEvalPack code pairs, orange: other). A judge that follows the criterion would place points in the lower-right quadrant; all four judges show a positive slope.

reversal share is 0.000 for all four judges. In the length control, on 72 pairs with different character lengths, the longer response is picked in both orders in 0.000, 0.000, 0.014 and 0.250 of pairs (random-judge level 0.25). These controls use the argmax readout and inherit its letter-bias problem; we did not re-analyze them with the calibrated readout. RewardBench sections are unbalanced (chat 12, chat-hard 65, reasoning 223 pairs); per-section tables are in `results/analysis.md`.

VI. DISCUSSION AND LIMITATIONS

Interpretation. On this sample, the pair-dependent part of the four judges’ A-minus-B logit barely depends on whether they are asked which is better or which is worse; the question wording mostly moves the letter prior. This is criterion-blindness in the sense that whichever response the judge leans towards is insensitive to the criterion; for three of the four judges that lean is close to chance with respect to the

RewardBench label. Our earlier, pre-registered interpretation that the smallest judges are “position-locked” and that the negation test is therefore uninformative for them was wrong: it followed from an argmax readout that a large letter offset dominates. The positive correlation may reflect any pair-level feature the judges respond to (quality, length, surface form); d cannot say which, only that it is not inverted by the negated question. For practitioners, the lesson is methodological as much as substantive: argmax verdicts of letter-biased judges can hide graded preferences, and an order-averaged logit difference is a cheap check that changes the diagnosis.

Scope. The claim is limited to four Qwen judges up to 1.7B, a next-token argmax A/B readout (and its order-averaged logit difference), thinking disabled, one prompt template, and a 300-pair RewardBench subsample in which 217 pairs are HumanEvalPack code pairs and responses are at most 150 tokens. We make no claim about larger judges, reasoning-

enabled judges [5], or generated verdicts.

Threats to validity. (i) *Post-hoc analysis*: the calibrated readout, the letter-biased null and the subset analyses were added after review; they were not pre-registered. Their CIs are descriptive, not tests with multiplicity control. (ii) *Calibration model*: d removes a constant additive letter offset; order effects that interact with content are not removed. Content-by-slot interactions (e.g. an “A” logit that depends on the length of the text in slot A) enter d identically under both questions and would push r upward without any criterion-independent preference for a response. (iii) *What d measures*: d is a pair-dependent shift that the question mostly leaves unchanged, not necessarily a quality judgment. Calibrated “better” accuracy is near chance (0.50–0.61) for three judges; only Qwen3-1.7B (0.81) shows clear content signal. For Qwen2.5-0.5B, d is tiny relative to the letter offset (median $|d_{\text{better}}|$ 0.047 vs. mean offset 2.80), so its $r = 0.95$ is not evidence of a coherent preference. (iv) *Readout*: no generation-based check was run; for Qwen3-1.7B the A/B mass is low on a quarter of passes, and the mass >0.9 subset ($n = 140$) is a selected sample. (v) *Sample*: code-heavy, near-identical HEP pairs; the non-HEP subset is small ($n = 83$); chat has 12 pairs. (vi) *Hypotheses*: H1 could not be evaluated as designed (tiny conditioning subsets); H2 is supported as pre-registered but, given the letter rates, nearly tautological. (vii) *Protocol deviation*: the development-set mass check was run for one judge only. (viii) *Novelty*: the related-work scan used arXiv and web search only.

What was not done. No judge of 3–4B parameters and no thinking-enabled condition were run (budget rule; see Section IV); no generation-based readout; no calibrated re-analysis of the W2 and length controls; no prompt-wording search beyond W1/W2.

VII. CONCLUSION

Asked “which response is worse?”, four Qwen judges between 0.5B and 1.7B parameters (thinking off, A/B logit readout) almost never reverse their “better” verdict under argmax (0, 0, 0 and 1 of 300 pairs). The pre-registered diagnosis of position-locking does not hold up: it is largely implied by the judges’ letter rates. After removing the letter offset, the pair-dependent logit shift under “worse” is strongly positively correlated with that under “better” ($r = 0.63$ to 0.90 for the three judges that do not always answer “A”; 0.95 for Qwen2.5-0.5B, whose shift is tiny relative to its letter offset), including on non-code pairs and high-mass pairs, so these judges are criterion-blind on this sample. Only Qwen3-1.7B shows clear content signal under “better” (calibrated accuracy 0.81). Whether generated verdicts, thinking, or larger judges change this is open.

REPRODUCIBILITY STATEMENT

Repository path: `projects/small-judge-reversal/`. Raw per-pass outputs: `results/raw_<model>_r<k>.json`; model and data revisions: `results/models.md`; environment: `results/pip_freeze.txt`; commands: `results/commands.md`. From the project directory:

```
uv venv .venv --python 3.14
uv pip install --python .venv/bin/python \
  -r requirements.txt
./fetch_data.sh
OMP_NUM_THREADS=4 .venv/bin/python -I \
  src/run.py prepare
# for M in the 4 models, s in 0 1 2:
OMP_NUM_THREADS=4 .venv/bin/python -I \
  src/run.py run --model $M --resample $s
.venv/bin/python -I src/analyze.py
.venv/bin/python -I src/calibrated.py
# venv with paper/requirements-paper.txt:
python -I paper/make_figures.py
```

Only `src/analyze.py`, `src/calibrated.py` and `paper/make_figures.py` are needed to re-derive all tables and figures from the committed raw outputs.

AI-AUTHORSHIP DISCLOSURE

Produced by an autonomous AI agent (Claude) on behalf of @pranavnandyalam. Not peer reviewed. The study design, code, experiments, analysis and this text were produced by the agent; the plan, results and a first version of this paper were reviewed by automated overseer, ethics and referee agents, not by human peer review.

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